

Output Content (OC)

Score: 3/5 — Moderate gaps | Confidence: medium

Benchmark: LAILA | Context: Arab-Country G10–12 Arabic Teachers: AI Essay Grading Assistance

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Annotation procedures are well-documented and rigorous: 6 annotators and 3 supervisors, all Arabic language teachers/lecturers with five holding MSc/PhD in Arabic [Q30, Q31]; structured training [Q102]; comprehensive guidebook with exemplars [Q41]; annotator blinding to student identity [Q42]; double scoring with mean-rounded-down aggregation and supervisor escalation at $HOL \geq 6$ [Q43]; QWK IAA of 0.66–0.75 classified as substantial [Q44]; fair compensation [Q101]. These are real strengths. The validity concern is that all annotators appear to be Qatar-based and trained on QUTC CAST [Q30, Q33], with no documentation of nationality diversity (gap_id 5: no documentation found). The elicitation marks OC as MODERATE because the user judges pan-Arab MSA standards uniform for surface linguistic quality (Q1), which lowers — but does not eliminate — the risk that evaluative judgments on Style, Development, and Organization carry implicit Gulf-curriculum norms. The aggregation method (mean-rounded-down) [Q43] could erase minority interpretations on borderline cases. P5 had a 23.3% third-annotator escalation rate [Q45], suggesting some prompts produced systematic annotator disagreement. Overall, methodologically strong but with an unresolved annotator-pool diversity question that matters for cross-country deployment.

ID	Evidence
Q1	"LAILA, the largest publicly available Arabic AES dataset to date, comprising 7,859 essays annotated with holistic and trait-specific scores on seven dimensions: relevance, organization, vocabulary, style, development, mechanics, and grammar." (p.1)
Q30	"The hired annotation team consisted of 6 annotators and 3 supervisors, all of whom were Arabic language teachers or lecturers with educational backgrounds." (p.5)
Q31	"Five members of the team hold advanced degrees (MSc or PhD) in the Arabic language." (p.5)
Q33	"All essays in LAILA dataset were scored using the Core Academic Skills Test (CAST) rubric [D7], developed by the Qatar University Testing Center (QUTC)." (p.5)
Q41	"To ensure consistency, 2 supervisors developed an annotation guidebook with detailed terminology, exemplars, and annotated practices for each prompt type. Annotators were required to review these materials and complete training sessions before formal annotation. Moderation sessions followed, where discrepancies were discussed and interpretations of the rubric were harmonized." (p.6)
Q42	"The scoring process was conducted using the Assessment Gourmet Platform, which supports large-scale, anonymized essay scoring. The platform ensured that annotators were blinded to student identity [D6], while only supervisors had access to annotator metadata for monitoring purposes." (p.6)
Q43	"The essays were randomly distributed among the annotators to minimize systematic bias, and the scoring sessions were capped to prevent annotators' fatigue. Each essay was independently scored by two annotators across all traits. If the difference in HOL scores between the two annotators was less than 6 points, the mean of the two scores was computed and then rounded down to the nearest integer; this rounded value was adopted as the final score for each trait. However, essays with large discrepancies between annotators (≥ 6 points difference in the HOL score) were flagged and escalated to a supervisor (a third annotator in this case), who provided the final adjudicated score and offered feedback to the annotators to improve alignment and consistency in subsequent batches of essays." (p.6)
Q44	"We compute IAA using Quadratic Weighted Kappa (QWK) (Williamson et al., 2012) to assess agreement between the two main annotators (A1 and A2), prior to adjudication. Table 4 reports QWK per trait and prompt, with agreement strength following Landis and Koch (1977). Overall, IAA was substantial across prompts, with an average QWK ranging from 0.66 (P5) to 0.75 (P1), showing strong consistency." (p.6)
Q45	"However, variability emerges by prompt and trait. In particular, P5 showed the lowest average agreement, noting that it has the highest percentage of essays that require a third annotator (A3: 23.3%; Table 3). This suggests greater initial disagreement" (p.6)

Q101	"All annotators were qualified Arabic language educators who received fair compensation at or above local professional rates." (p.10)
Q102	"They completed structured training on annotation guidelines, and inter-annotator disagreements were resolved through systematic adjudication procedures." (p.10)

Strengths

- Annotators are qualified Arabic-language educators, five with advanced degrees [Q30, Q31]
- Structured training, annotation guidebook, and moderation sessions [Q41, Q102]
- Double annotation with explicit adjudication procedure for disagreements ≥ 6 HOL points [Q43]
- Substantial inter-annotator agreement (QWK 0.66–0.75) [Q44]
- Annotator blinding to student identity reduces demographic bias [Q42, Q98]
- Fair compensation at or above local professional rates [Q101]
- IRB approval and informed consent documented [Q9, Q95]

ID	Evidence
Q9	"IRB Number: QU-IRB 159/2024-EA" (p.1)
Q30	"The hired annotation team consisted of 6 annotators and 3 supervisors, all of whom were Arabic language teachers or lecturers with educational backgrounds." (p.5)
Q31	"Five members of the team hold advanced degrees (MSc or PhD) in the Arabic language." (p.5)
Q41	"To ensure consistency, 2 supervisors developed an annotation guidebook with detailed terminology, exemplars, and annotated practices for each prompt type. Annotators were required to review these materials and complete training sessions before formal annotation. Moderation sessions followed, where discrepancies were discussed and interpretations of the rubric were harmonized." (p.6)
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Q95	"The collection of LAILA received approval from an Institutional Review Board (IRB Number: QU-IRB 159/2024-EA) and was conducted with informed consent from both students and their legal guardians." (p.10)
Q98	"Annotation was conducted on a platform that prevented annotators from accessing student identities or demographic information." (p.10)
Q101	"All annotators were qualified Arabic language educators who received fair compensation at or above local professional rates." (p.10)
Q102	"They completed structured training on annotation guidelines, and inter-annotator disagreements were resolved through systematic adjudication procedures." (p.10)

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement

OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: Partially. Ground-truth labels reflect qualified Arabic-educator judgment [Q30] but specifically Qatar-based educators trained on QUTC CAST [Q33]; whether they reflect Egyptian, Levantine, or Maghrebi educator perspectives is undocumented.

OC-2: Potential disagreement is plausible on Style, Development, and Organization given different national rhetorical traditions; the elicitation flags this as warranting verification (Q1). The pan-Arab MSA assumption should hold for surface mechanics/grammar but is less certain for evaluative traits.

OC-3: Annotator demographics are partially documented — qualifications and degree level [Q30, Q31] are reported, but nationality, dialect background, and curricular training origin beyond CAST are not. No Datasheet or Data Statement section specifies these.

OC-4: INSUFFICIENT DOCUMENTATION — no re-annotation by representative non-Qatari Arab annotators is reported; the LAILA paper does not include any cross-country annotator validation study.

OC-5: Aggregation is mean-of-two-scores rounded down to nearest integer [Q43], with supervisor adjudication only when HOL discrepancy ≥ 6 points. This may erase legitimate minority perspectives on borderline cases (e.g., a 3 vs. 4 disagreement is averaged and rounded down to 3 without further review).

OC-6: Documented label-correctness considerations: (1) annotator pool nationality undocumented [gap_id 5]; (2) substantial but not perfect IAA, with P5 showing 23.3% escalation [Q45]; (3) round-down aggregation may bias scores slightly downward on disagreement cases [Q43].

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WEB-5	Egypt's holistic grading culture suggests Egyptian teachers may apply different evaluative standards than Qatar-based annotators (https://grokipedia.com/page/academic_grading_in_egypt)

Information Gaps

- Annotator nationality and dialectal background distribution (gap_id 5: no documentation found)
- Whether the round-down aggregation introduces systematic downward bias on borderline scores
- What proportion of supervisor adjudications involved trait-specific (vs. holistic) disagreements

Requires Expert Verification

- Whether non-Qatari Arab Arabic-language educators would assign systematically different Style, Development, or Organization scores on the same essays
 - Inquiry to LAILA authors regarding annotator nationality composition
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Remediation

Gap 1: Annotator nationality diversity within the Qatar-based pool is undocumented [gap_id 5]; cross-country annotator agreement is unknown

Recommendation: Direct inquiry to LAILA authors for annotator nationality composition; commission a small cross-country re-annotation study (~500 essays) with Egyptian, Saudi, Lebanese, and Moroccan Arabic-language educators to quantify QWK drift on Style, Development, and Organization.